23A03605b	**INDUSTRIAL SAFETY** **(Professional Elective-III)**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

Course objectives: The objectives of the course are to	
1	Understand the concepts of industrial safety and management.
2	Demonstrate the accident preventions and protective equipment.
3	Understand and apply the knowledge of safety acts
4	knowledge about fire prevention and protection systems
5	Understand and apply fire safety principles in buildings

COURSE OUTCOMES On successful completion of this course the student will be able to		
1	Students learn the concepts of industrial safety and management.	*L2*
2	Learn about the smart machines and smart sensors	L1,L2
3	Apply IoT to Industry 4.0 and they are able to make a system tailor-made as per requirement of the industry	L4,L5
4	Students learn about fire prevention and protection systems.	L2,L3
5	Students learn and apply the fire safety principles in buildings	L2,L4

UNIT-I

INTRODUCTION TO THE DEVELOPMENT OF INDUSTRIAL SAFETY AND MANAGEMENT: History and development of Industrial safety: Implementation of factories act, Safety and productivity, Safety organizations. Safety committees and structure, Role of management and role of Govt.in industrial safety.

UNIT-II

ACCIDENT PREVENTIONS AND PROTECTIVE EQUIPMENT: Personal protective equipment, Survey the plant for locations, Part of body to be protected, Education and training in safety, Prevention causes and cost of accident, Housekeeping, First aid, Accident reporting, Investigations. Industrial psychology in accident prevention, Safety trials, Safety related to operations.

UNIT-III

SAFETY ACTS: Features of Factory Act, Introduction of Explosive Act, Boiler Act, ESI Act, Workman's compensation Act, Industrial hygiene, Occupational safety, Diseases prevention, Ergonomics, Occupational

diseases, stress, fatigue, health, safety and the physical environment, Engineering methods of controlling chemical hazards, safety and the physical environment, Control of industrial noise and protection against it, Code and regulations for worker safety and health, codes for safety of systems.

UNIT-IV

FIRE PREVENTION AND PROTECTION: Sources of ignition – fire triangle – principles of fire extinguishing – active and passive fire protection systems – various classes of fires – A, B, C, D, E-Fire extinguishing agents- Water, Foam, Dry chemical powder, Carbon-dioxide Halon alternatives Halocarbon compounds-Inert gases, dry powders – types of fire extinguishers – fire stoppers –hydrant pipes – hoses – monitors – fire watchers – layout of stand pipes – fire station-fire alarms and sirens – maintenance of fire trucks – foam generators – escape from fire rescue operations – fire drills –first aid for burns.

UNIT-V

BUILDING FIRE SAFETY: Objectives of fire safe building design, Fire load, fire resistant material and fire testing – structural fire protection – structural integrity – concept of egress design -exit– width calculations – fire certificates – fire safety requirements for high rise buildings.

TEXT BOOKS:

1. Occupational Safety Management and Engineering Willie Hammer–PrenticeHall (2000)
2. Purandare D.D & Abhay D.Purandare, "Handbook on Industrial Fire Safety" P&A publications, NewDelhi, 2006.

REFERENCE BOOKS:

1. Installation, Servicing and Maintenance Bhattacharya, S.N.-S.Chandand Co.
2. Jain VK "Fire Safety in Building" New Age International 1996.
3. Reliability, Maintenance and Safety Engineering by Dr.A. K.Guptha
4. A Text book of Reliability and Maintenance Engineering by Alakesh Manna
5. McElroy,FrankE.,"Accident Prevention Manual for Industrial Operations", NSC, Chicago, 1988.
6. Green, A.E.,"High Risk Safety Technology", John Wiley and Sons, 1984.

Online Learning Resources:

https://nptel.ac.in/courses/110105094

https://youtube.com/playlist?list=PLbRMhDVUMngdXebaRB59KdKwstzuAovua&si=FcbDQzZK6i_3TASD

https://youtube.com/playlist?list=PLbRMhDVUMngdXebaRB59KdKwstzuAovua&si=6RaMiYhEkp5-EfAH

https://youtube.com/playlist?list=PLln3BHg93SQ8RYKhe9czOHq1hVjpEWMts&si=5y0WMqX3wrvvispq